Will Ye

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Summary

Expert in building scalable AI/ML solutions for complex problems.

Skills

Languages: Python, SQL, TypeScript, C/C++
Libraries/APIs: Scikit-learn, Pandas, Pytorch, Tensorflow, SpaCy, XGBoost, OpenCV, Natural Language Processing (NLP), Hugging Face
Storage: Neo4j, Redshift, Graph Databases, Elasticsearch
Platforms: Linux/Unix, AWS, Docker, Kubernetes, Google Cloud Platform, Microsoft Azure

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Experience

Ramp — **Remote**
Senior Software Engineer (Applied AI, Founding Member) — *05/2023 - Present*

- Launched LLM-powered expense and receipt automation integrated with Ramp's corporate card and expense workflows; reduced manual receipt matching time by 65% and increased auto-approval coverage by 30%.
- Designed RAG system for policy-aware expense explanations using customers' policy docs and historical decisions; implemented vector store with hierarchical chunking, hybrid (BM25 + dense) retrieval, and query routing.
- Built multi-model routing layer (OpenAI, Anthropic, Cohere, internal finetunes) leveraging cost/latency/accuracy trade-offs with dynamic fallbacks; cut average per-request cost by 28% while improving success rate 4.2pp.
- Introduced a robust evaluation harness (Ragas + custom judges + human review) for policy compliance, factuality, and actionability; established "eval gates" in CI to prevent regressions on prompts and models.
- Productionized tool/function calling for finance automation tasks (categorization, memo generation, policy checks, merchant normalization); stabilized via structured output (JSON schema), deterministic parsing, and retries.
- Led safety and compliance controls: PII redaction pre/post-processing, rate-limited function calls, audit logs, and model access governance; collaborated with Risk/Legal to meet SOC2 and customer procurement requirements.
- Implemented lightweight fine-tuning/LoRA adapters for domain-specific reasoning and terminology; delivered 9–12% accuracy lift on policy application tasks versus base models at the same temperature.
- Built prompt/version registry with feature flags, telemetry, and reversible rollouts; enabled rapid experimentation and safe hotfixes across services.
- Optimized serving stack using vLLM and KV cache reuse; achieved p95 latency under 900ms for on-platform completions with streaming UI.
- Mentored engineers and partnered with Product/Design to build explainable outputs and escalation flows; shipped features that contributed to a measurable lift in user adoption and NPS among finance admins.

Cohere — **Remote**
Senior Software Engineer (First Employee; acquired by Ramp) — *05/2021 - 05/2023*

- First engineering hire building NLP model capabilities and developer platform; contributed to core text-embedding and generation offerings used by early enterprise customers.
- Co-developed high-quality embedding pipelines (multi-lingual, domain-adapted) and evaluated with MTEB-style suites; delivered 12–18% improvements on retrieval tasks for finance and support domains.

- Implemented retrieval-augmented generation reference stack (indexing, chunking, reranking with cross-encoders) for customer POCs; materially improved grounded answer accuracy and reduced hallucinations.
- Built performant serving infrastructure (Triton + PyTorch + quantization-aware exports) and autotuning for throughput/latency; enabled 2–3x QPS improvements with minimal quality loss using 8-bit/4-bit quantization.
- Created evaluation tooling for instruction-following and safety using synthetic datasets, pairwise preference comparisons, and human-in-the-loop annotation; supported early DPO-style preference optimization.
- Implemented guardrails (regex/policy filters, classifier gating, sensitive-topic detectors) and prompt hardening; reduced unsafe output rate by >60% in internal red-teaming.
- Designed dataset curation pipelines: deduplication, contamination checks, PII scrubbing, and quality scoring; improved downstream model generalization and reduced leakage.
- Built SDK improvements and sample apps for search, Q&A, and summarization; collaborated with Solutions to accelerate pilot conversions.
- Contributed to internal tooling for LoRA/PEFT finetunes and experiment tracking; standardized metrics and artifact lineage across teams.

Amazon Web Services **Remote**
Software Engineer *05/2020 - 05/2021*

- First engineer on an internal tooling team; built services that automated developer workflows for large-scale distributed systems.
- Implemented metadata indexing and semantic search over internal docs using early embedding models; accelerated on-call resolution and reduced mean time to mitigation.
- Delivered high-reliability APIs with canary deploys, CloudWatch metrics, and error budgets; maintained 99.9%+ service availability.
- Drove integration with AWS primitives (S3, Lambda, Step Functions, DynamoDB) to orchestrate document ingestion, deduplication, and access controls at scale.

Capital One **Remote**
Software Engineer Intern *06/2019 - 08/2019*

- Prototyped a credit risk feature store for model features (transaction aggregates, bureau-derived features) with lineage and monitoring; informed subsequent platform design.
- Built dashboards and batch pipelines (Spark + Airflow) to monitor feature drift and data freshness for underwriting models.

Education

Duke University **04/2016 - 05/2020**
Bachelor's, Computer Science